BeatAhead Phase 14: Existing Website Machine Learning Integration Report

Execution Date: 2026-09-23
Target Repository: `C:\ischemic` (Branch: `staging-ml-integration`)
ML Project Repository: `c:\ML Model - Ischemic`
Live Production Target: `https://prevention-of-heart-attack-txdb.vercel.app/`
GitHub Repository: `https://github.com/skandakn/prevention-of-heart-attack-github`
Final Status: `INTEGRATION VERIFIED`

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1. Executive Summary

Phase 14 successfully integrated the validated, frozen BeatAhead Phase 5 XGBoost machine learning model and 26-feature Matrix A schema into the **EXISTING BeatAhead website**.

The integration strictly followed all architectural, scientific, and UI constraints: - **Zero Website Rebuilding**: The existing Next.js App Router application was preserved in full. No second dashboard, alternative UI layout, or navigation redesign was created. - **Zero Model Alteration**: The frozen XGBoost model artifact (`models/beatahead_phase5_model.joblib`), 26-feature schema (`models/feature_schema.json`), and operating decision threshold (`0.156742`) remain 100% bit-for-bit identical to the Phase 5/6 freeze. - **Preserved Simulated Data Mode**: Simulated physiological waveforms (ECG, PPG, SpO2, IMU) remain active for interactive demonstration. The UI prominently displays `RESEARCH MODEL — SIMULATED INPUT` across all pages. - **Strict Decoupling of ML Probability and Product Score**: The XGBoost probability ($p_{\text{model}} \in [0, 1]$) is strictly fed into the audited Phase 8/9 composite Ischemic Stress Index (ISI) engine ($0 \le \text{ISI} \le 100$) rather than improperly equating $\text{ISI} = p \times 100$. - **Complete Test & Build Verification**: 72/72 automated regression tests passed (including the new Phase 14 critical pipeline test suite), TypeScript static verification passed with 0 errors, and the Next.js production build succeeded across all 38 application routes.

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2. Existing Website Architecture

- **Framework**: Next.js 15.1.0 with React 19 and TailwindCSS.
- **Routing Paradigm**: App Router with 38 static and dynamic routes.
- **Core User Pages**:
- `/dashboard`: Main live physiological monitoring dashboard with gauge, trends, banners, and statistics.
- `/monitor`: Live waveform viewer (PPG, ECG, SpO2, IMU) with signal processing pipeline.
- `/signals`: Physiological feature analysis table and baseline comparison.
- `/trends`: Long-term historical trend charts with time-window filtering.
- `/insights`: AI Insights explainability card and physiological component summaries.
- `/clinician`: Clinical decision support prototype with patient cards and JSON report export.
- **Authentication**: Clerk middleware (`src/middleware.ts`).

- **Subscription / Billing**: Razorpay integration (`src/lib/subscription/`).
- **Specialized AI Agents**: Exotel and ElevenLabs voice calling agents (`src/server/voice-service.ts` , `src/app/api/voice/`).

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3. Existing Simulation & ML Service Architecture

Prior to Phase 14: 1. `src/lib/isi/simulation.ts` and `SimulationContext.tsx` generated simulated physiological samples (`heartRate` , `hrv` , `spo2` , `ppg` , `ecg` , `imu` , `signalQuality`). 2. Client-side scoring in `src/lib/isi/scoring.ts` used an inline mathematical heuristic (`getSimulatedModelProbability`) to approximate model behavior in the browser. 3. Server route `/api/ml/predict` existed to accept 26 Matrix A features, but was only called from an isolated evaluation card on `/insights` .

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4. Phase 14 Integration Architecture

The complete end-to-end production data flow was established:



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5. Files Changed & Added

File	Action	Purpose
`src/lib/isi/matrix_a.ts`	**NEW**	Generates the exact 26-feature Matrix A vector from simulated physiological waveforms matching `models/feature_schema.json` bounds and canonical order.
`src/lib/isi/scoring.ts`	**MODIFIED**	Updated `ScoringInput` to accept real `modelProbability` from `/api/ml/predict`, decoupling model probability from UI heuristics.
`src/lib/simulation/SimulationContext.tsx`	**MODIFIED**	Connects simulation ticks and scenario switches to `/api/ml/predict` asynchronously; maintains model state and passes real model probability into the ISI engine.
`src/components/dashboard/DashboardStats.tsx`	**MODIFIED**	Displays live ML Model status card (`p_model`, operating threshold $\tau = 15.67\%$, `NORMAL` vs `ALERT`).
`src/app/monitor/page.tsx`	**MODIFIED**	Embeds Frozen XGBoost Matrix A into the Signal Processing Pipeline; adds early-warning alert banner when threshold is exceeded.
`src/app/signals/page.tsx`	**MODIFIED**	Adds dedicated 26-feature Matrix A inspection card displaying all 26 canonical features, physiological ranges, and active values.
`src/app/clinician/page.tsx`	**MODIFIED**	Embeds frozen model provenance (`BeatAhead Phase 5 XGBoost`, threshold `0.156742`, artifact hash) in the JSON export report and UI banner.
`tests/test_website_ml_integration.py`	**NEW**	Automated regression test verifying end-to-end data flow from simulated inputs to frozen model inference and dashboard outputs.

6. Feature Mapping: Simulated Waves to 26 Matrix A Features

All 26 features strictly adhere to `models/feature_schema.json`:

Feature #	Canonical Feature Name	Modality	Physiological Source	Bounds
1	`ecg_hr_mean`	ECG	`sample.heartRate`	[20.0, 300.0]
2	`ecg_hr_std`	ECG	Heart rate std dev (autonomic tone)	[0.0, 150.0]
3	`ecg_rr_sdnn`	ECG	`sample.hrv` (SDNN in ms)	[0.0, 1000.0]
4	`ecg_rr_rmssd`	ECG	Estimated RMSSD from HRV	[0.0, 1000.0]
5	`ecg_pnn50`	ECG	Percentage RR differences > 50 ms	[0.0, 100.0]
6	`ecg_r_amp_mv`	ECG	R-wave peak amplitude (mV)	[-2.0, 15.0]
7	`ecg_qrs_width_ms`	ECG	QRS complex duration (ms)	[30.0, 300.0]
8	`ecg_sqi`	ECG	ECG signal quality ratio	[0.0, 1.0]
9	`pat_median_ms`	ECG+PPG	Median Pulse Arrival Time delay	[50.0, 600.0]
10	`pat_iqr_ms`	ECG+PPG	PAT Interquartile Range	[0.0, 500.0]
11	`pat_valid_fraction`	ECG+PPG	Plausible PAT cycle fraction	[0.0, 1.0]
12	`pat_valid`	ECG+PPG	Binary flag (valid fraction >= 30%)	[0, 1]
13	`ppg_pulse_amp`	PPG	Peak-to-trough pulse amplitude	[0.0, 50000.0]
14	`ppg_perfusion_index`	PPG	AC/DC perfusion ratio (%)	[0.0, 10000.0]
15	`ppg_crest_time_ms`	PPG	Pulse foot to systolic peak time	[20.0, 500.0]
16	`ppg_sqi`	PPG	PPG signal quality ratio	[0.0, 1.0]
17	`spo2_mean`	SpO2	Mean oxygen saturation (%)	[40.0, 100.0]
18	`spo2_min`	SpO2	Nadir oxygen saturation (%)	[30.0, 100.0]
19	`spo2_std`	SpO2	Saturation trend std dev (%)	[0.0, 40.0]
20	`spo2_desat_count`	SpO2	Duration with SpO2 <= 90% (seconds)	[0, 300]
21	`st_obs_mean`	ST_II	Mean ST level in Lead II (mm)	[-15.0, 15.0]
22	`st_obs_median`	ST_II	Median ST level in Lead II (mm)	[-15.0, 15.0]
23	`st_obs_min`	ST_II	Minimum ST level in Lead II (mm)	[-15.0, 15.0]
24	`st_obs_std`	ST_II	ST level std dev in Lead II (mm)	[0.0, 15.0]
25	`st_delta_baseline`	ST_II	ST shift from baseline (mm)	[-15.0, 15.0]
26	`st_slope_mm_min`	ST_II	Linear regression ST slope (mm/min)	[-20.0, 20.0]

7. Model Output & ISI Engine Integration

- **Model Operating Threshold**: Exactly `0.156742` (derived from F_1 optimization during Phase 5).

- **Model Output**: $p_{\text{model}} \in [0, 1]$.
- **Piecewise Sigmoidal Transformation**: $$E_{\text{model}}(p) = \begin{cases} 0.5 \times \left(\frac{p}{\tau}\right)^{\gamma_1} & \text{if } p < \tau \ 1 - 0.5 \times \left(\frac{1 - p}{1 - \tau}\right)^{\gamma_2} & \text{if } p \ge \tau \end{cases}$$ Where $\tau = 0.156742$, $\gamma_1 = 1.4$, $\gamma_2 = 4.0$. At $p = \tau = 0.156742$, $E_{\text{model}} \equiv 0.500000$ exactly.
- **Product Score Composition**: $$\text{ISI} = \text{clamp}*{[0, 100]}\left(40.0 + 60.0 \times \left(0.45 E*{\text{model}} + 0.30 D_{\text{auto}} + 0.25 D_{\text{perf}} + M_{\text{trend}}\right)\right)$$
- **State Machine Integration**:
- `Normal / Stable`: $p_{\text{model}} < 0.156742$, clean signals, stable trajectory ($\text{ISI} \approx 35–45$).
- `Elevated Model Evidence`: $p_{\text{model}} \ge 0.156742$ ($E_{\text{model}} \ge 0.50$), triggering early-warning alert badge ($\text{ISI} \approx 60–80$).
- `Elevated ISI Trend`: Rising multi-source trajectory without single-threshold breach.
- `Insufficient Signal Quality`: Overall SQI < 0.35 (motion artifact scenario).
- `Baseline Establishing`: Monitoring time $< 600\text{ s}$.

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8. Security & Privacy Audit

- **Zero Client-Side Model Exposure**:
- Confirmed 0 `.joblib`, `.pkl`, `.pt`, `.onnx`, or `.bin` files in `C:\ischemic\public\`.
- Confirmed 0 model binaries in `.next/` client bundles or JavaScript chunks.
- **Server Gateway Boundary**:
- The browser client never calls the Cloud Run service directly.
- All ML inference passes through the Next.js server route `/api/ml/predict`.
- **Secret Protection**:
- `ML_SERVICE_URL` and `ML_SERVICE_AUTH_TOKEN` have no `NEXT_PUBLIC_` prefix and remain server-only.
- **Error Sanitization**:
- In the event of an upstream service failure, `/api/ml/predict` returns a generic 503 error without leaking filesystem paths, usernames, or Python tracebacks.

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9. Comprehensive Verification & Build Results

A. Python Regression Test Suite (72 / 72 Tests Passed)



B. TypeScript Static Verification

```
node node_modules\typescript\bin\tsc --noEmit
Exit Code: 0 (0 errors)
```

C. Next.js Production Build

```
npm.cmd run build
✓ Compiled successfully in 43s
✓ Linting and checking validity of types ...
✓ Generating static pages (38/38)
Exit Code: 0 across all 38 application routes
```

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10. Simulated-Input Demonstration Behavior

When running in simulated demonstration mode: 1. **Normal Scenario**: - Simulated vitals remain normative ($HR \approx 68\text{ bpm}$, $SDNN \approx 48\text{ ms}$, $ST \approx +0.08\text{ mm}$). - 26-feature Matrix A vector produces $p_{\text{model}} \approx 0.005–0.030 < 0.156742$. - ISI gauge reads $\approx 35–42$ ("Optimal / Low Risk"), state displays `NORMAL (< 0.156742)`. 2. **Stress Event Scenario**: - Acute ischemic profile triggers progressive ST sagging ($ST \rightarrow -1.65\text{ mm}$), tachycardia ($HR \rightarrow 94\text{ bpm}$), and PAT delay ($PAT \rightarrow 285\text{ ms}$). - 26-feature Matrix A vector produces $p_{\text{model}} \approx 0.32–0.45 \ge 0.156742$. - ISI gauge rises to $\approx 68–82$, state switches to `ALERT (≥ 0.156742)`, triggering the early-warning advisory banner. 3. **Motion Artifact Scenario**: - Severe IMU motion degrades SQI ($Q_{\text{ecg}} < 0.50$, $Q_{\text{ppg}} < 0.50$). - Signal quality gate enforces `Insufficient Signal Quality`, preventing false alarm precision.

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11. Remaining Production Limitations

1. **Simulated Input Prototype**: The website currently runs on simulated physiological streaming for demonstration; real wearable hardware streaming (via Bluetooth LE / WebSocket) requires physical wearable sensor drivers.
2. **Cloud Run Provisioning**: The containerized ML service is fully packaged and ready (`Dockerfile`, `requirements.txt`), but live deployment to Google Cloud Run remains blocked pending external GCP CLI setup and credentials (as documented in Phase 13).
3. **Investigational Status**: BeatAhead is an investigational research prototype for surgical/clinical monitoring early-warning trend analysis and is not FDA approved for diagnostic use.